

# MATH 3551 A — Abstract Algebra I

## Practice Midterm 1 — SOLUTIONS

Taylor Dupuy, University of Vermont • Fall 2026

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**About this practice exam.** It was drafted with the help of Claude Opus 5, running in a Cowork harness, working from previous exams in this course. **It has not yet been timed or proofread.** Like the real exam, it is built to take well under the time you are given: about thirty minutes of a fifty-minute midterm period, and about a hundred minutes of the final's two hours forty-five. The spare time is deliberate. *Please hunt for mistakes.* If you find a wrong answer, an ambiguous or unanswerable problem, a problem that leans on something we have not covered, or one that takes far longer than its point value suggests, tell me — in class, in office hours, or by email. I would much rather hear it from you now than print it on the real exam.

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**Problem 1.** (30 points) **True or false; prove or disprove.**

- (a) **FALSE.** The order of a permutation is the lcm of its cycle lengths, so we would need a partition of at most 9 whose parts have lcm 18. Since  $18 = 2 \cdot 3^2$ , such a partition must contain a part divisible by 9 and a part divisible by 2; the smallest possibility is  $9 + 2 = 11 > 9$ . No such element exists.
- (b) **TRUE.**  $|xy| = |x||y|$  for rational  $x, y$ , and the operation on  $\mathbb{Q}^\times$  is multiplication, so  $\varphi(xy) = \varphi(x)\varphi(y)$ .
- (c) **TRUE.**  $D_{16} = D_{2n}$  with  $n = 8$ . No reflection is central (if  $sr^i$  were central then  $r(sr^i)r^{-1} = sr^{i-2} = sr^i$  forces  $r^2 = 1$ , false), and  $r^m$  is central iff  $r^m s = sr^m$ , i.e.  $r^m = r^{-m}$ , i.e.  $8 \mid 2m$ , i.e.  $m \in \{0, 4\}$ . So  $Z(D_{16}) = \{1, r^4\}$ .
- (d) **FALSE.**  $(\mathbb{Z}/15\mathbb{Z})^\times$  has order  $\varphi(15) = 8$ , but every element satisfies  $x^4 = 1$ : indeed  $\bar{2}^4 = \bar{16} = \bar{1}$ ,  $\bar{4}^2 = \bar{1}$ ,  $\bar{7}^4 = \bar{1}$ ,  $\bar{11}^2 = \bar{1}$ ,  $\bar{13}^4 = \bar{1}$ ,  $\bar{14}^2 = \bar{1}$ ,  $\bar{8}^4 = \bar{1}$ . So there is no element of order 8.
- (e) **TRUE.** Choose the 3 moved points in  $\binom{6}{3} = 20$  ways and arrange them in a cycle in  $(3-1)! = 2$  ways:  $20 \cdot 2 = 40$ .
- (f) **FALSE.**  $(\mathbb{Z}/8\mathbb{Z})^\times = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$  and each of  $\bar{3}, \bar{5}, \bar{7}$  squares to  $\bar{1}$ , so every non-identity element has order 2; but  $\mathbb{Z}/4\mathbb{Z}$  has an element of order 4. An isomorphism preserves orders of elements, so they are not isomorphic. (In fact  $(\mathbb{Z}/8\mathbb{Z})^\times \cong V_4$ .)

**Grading note.** Most of the credit here is in the *reason*, and the reason must be specific: a named counterexample, the actual permutation, or the actual count. A circled answer with no reason earns two of the five points; a wrong circle with a reason that is right about the underlying fact earns the three reason-points. Give full credit for a correct counterexample even if it is not the one printed above. The parenthetical remarks in each answer flag the error to expect on that item.

**Problem 2.** (25 points) **Inside  $\mathbb{Z}/72\mathbb{Z}$ .**

In  $\mathbb{Z}/72\mathbb{Z}$  the order of  $\bar{k}$  is  $72/\gcd(k, 72)$ , and  $\langle \bar{k} \rangle = \langle \bar{d} \rangle$  where  $d = \gcd(k, 72)$ .

(a)  $\gcd(30, 72) = 6$ , so  $|\overline{30}| = 72/6 = 12$ .

(b)  $\langle \overline{30} \rangle = \langle \overline{6} \rangle$ , the unique subgroup of order 12:

$$\{\overline{0}, \overline{6}, \overline{12}, \overline{18}, \overline{24}, \overline{30}, \overline{36}, \overline{42}, \overline{48}, \overline{54}, \overline{60}, \overline{66}\}.$$

(c)  $\varphi(72) = \varphi(2^3)\varphi(3^2) = 4 \cdot 6 = 24$ .

(d)  $\gcd(18, 72) = 18$ , so  $\langle \overline{18} \rangle$  is the subgroup of order  $72/18 = 4$ , namely  $\{\overline{0}, \overline{18}, \overline{36}, \overline{54}\}$ . Now  $\langle \bar{k} \rangle = \langle \overline{18} \rangle$  if and only if  $\bar{k}$  is one of the  $\varphi(4) = 2$  generators of that cyclic group of order 4, i.e.  $k \in \{18, 54\}$ . (Equivalently:  $\gcd(k, 72) = 18$ .)

**Grading note.** Award the points for the *method*:  $|\bar{k}| = n/\gcd(k, n)$  in (a),  $\langle \bar{k} \rangle = \langle \overline{\gcd(k, n)} \rangle$  in (b) and (d),  $\varphi(n)$  in (c). The expected error in (d) is to list all four elements of  $\langle \overline{18} \rangle$  rather than only its two generators; that is worth 4 of the 7 points. A student who answers (d) by testing all 72 values gets full credit if the answer is right.

**Problem 3.** (25 points) **Two permutations.**

(a)  $|\sigma| = \text{lcm}(3, 2, 4) = 12$ .

(b) Apply  $\tau$  first, then  $\sigma$ :  $1 \mapsto 2 \mapsto 5$ ,  $5 \mapsto 1 \mapsto 4$ ,  $4 \mapsto 5 \mapsto 2$ ,  $2 \mapsto 3 \mapsto 6$ ,  $6 \mapsto 6 \mapsto 8$ ,  $8 \mapsto 8 \mapsto 9$ ,  $9 \mapsto 9 \mapsto 3$ ,  $3 \mapsto 4 \mapsto 7$ ,  $7 \mapsto 7 \mapsto 1$ . Hence

$$\sigma\tau = (1\ 5\ 4\ 2\ 6\ 8\ 9\ 3\ 7),$$

a 9-cycle, so  $|\sigma\tau| = 9$ .

(c) The point 3 lies on the 4-cycle  $(3\ 6\ 8\ 9)$ , so  $\sigma^k$  fixes 3 if and only if  $4 \mid k$ . For  $0 \leq k < 12$  this gives  $k \in \{0, 4, 8\}$ .

(d)  $\sigma^{-1} = (1\ 7\ 4)(2\ 5)(3\ 9\ 8\ 6)$ .

**Grading note.** (b) is the part that takes time and it is where the composition order matters; a student who computes  $\tau\sigma$  instead gets  $(1\ 5\ 3\ 8\ 9\ 6\ 2\ 4\ 7)$  or similar — award 3 of the 6 points for a correct computation in the wrong order, provided the convention used is stated. In (c) the intended observation is that only the cycle containing 3 matters; students who compute all twelve powers of  $\sigma$  get full credit but will not have time for problem 5.

**Problem 4.** (20 points) **Centralizer and normalizer in the quaternion group.**

- (a)  $A = \langle i \rangle$ , so  $C_G(A) = C_G(i)$ . Certainly  $\pm 1$  (central) and  $\pm i$  (powers of  $i$ ) commute with  $i$ . On the other hand  $ji = -k$  while  $ij = k$ , so  $j$  does not commute with  $i$ ; the same computation applies to  $-j, k, -k$ . Hence

$$C_G(A) = \{1, -1, i, -i\} = A.$$

- (b)  $jij^{-1} = -i \in A$ , and  $j(-1)j^{-1} = -1 \in A$ , so  $jAj^{-1} = A$  and  $j \in N_G(A)$ . Since  $A \leq N_G(A)$  as well and  $|N_G(A)|$  divides 8 and is more than 4, we get  $N_G(A) = Q_8 = G$ . (Alternatively:  $|G : A| = 2$ , and every subgroup of index 2 is normal.)

- (c)  $|G : A| = 8/4 = 2$ , and the two left cosets are

$$A = \{1, -1, i, -i\}, \quad jA = \{j, -j, ji, -ji\} = \{j, -j, -k, k\}.$$

They are disjoint, each of size 4, and their union is  $G$ :  $2 \cdot 4 = 8 = |G|$ , as Lagrange requires.

**Grading note.** As in the real exam, the point is that  $C_G(A) \neq N_G(A)$ : the element  $j$  inverts  $i$  rather than fixing it. A student who reports  $C_G(A) = G$  “because  $Q_8$  is almost abelian” has missed the entire problem. Give full credit in (b) for the index-2 shortcut. In (c) accept  $\{j, -j, k, -k\}$  in any order; note  $ji = -k$ , so a student who writes  $jA = \{j, -j, k, -k\}$  without computing  $ji$  has still got the right set.